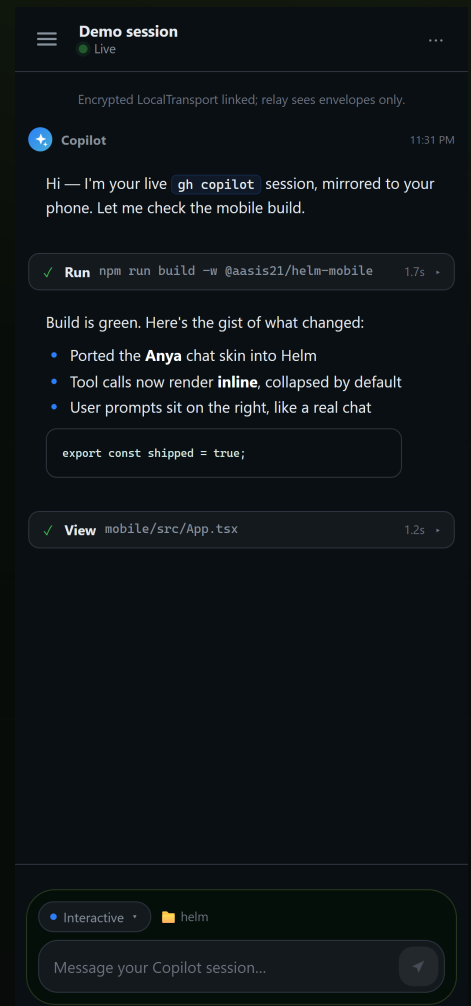




— PRODUCT DESIGN DOSSIER • CHAT SURFACE V2

A second screen for your *terminal*.

Helm binds to your live **gh copilot** session — watch the token stream, read the diffs, answer permission prompts, and steer the agent from your palm. **End-to-end encrypted**. Zero-persistence relay.

EDITION • JUNE 2026 SURFACE • [USEHELM.NETLIFY.APP](https://usehelm.netlify.app)

SOURCE • [GITHUB.COM/AASIS21/HELM](https://github.com/aasis21/helm)

— THE PROBLEM

Walk away from the desk.

Agent runs are long. Repo-wide refactors, **/fleet** commands, multi-step plans — they don't need you watching. Until, suddenly, they do: a file write to approve, a test to rerun, a wrong turn to stop.

Helm is the encrypted viewport that lets you **leave the terminal and still hold the controls** — without devtunnels, inbound firewall rules, or a background daemon eating your RAM.

— OPERATING PRINCIPLES

🔌 Ephemeral

The channel lives exactly as long as your **gh copilot** session. Close the terminal and it vanishes — no zombies, no background service.

🔒 Air-Gapped

A fresh **AES-256-GCM** key is derived on-device from a scanned QR. The relay only ever carries ciphertext.

⚙️ Native

A true CLI extension on the agent's **RPC loop** — not a scraper, not a wrapper. It sees events as they happen.

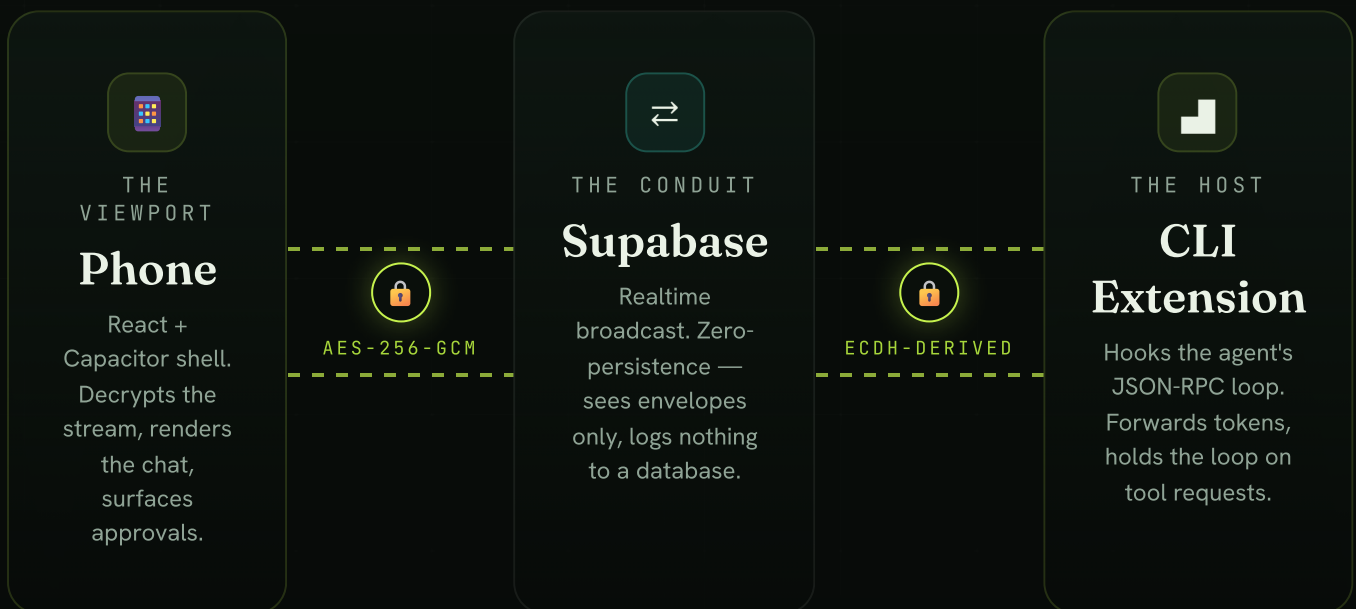
— CONTENTS

- 01 The Brief
- 02 System Architecture
- 03 The Interface
- 04 Multi-Session
- 05 Design System
- 06 Engineering & Assurance

— HOW IT CONNECTS

Three nodes, one encrypted seam.

Your phone never talks to the agent directly. It speaks to a volatile Supabase channel, which routes opaque envelopes to the extension riding inside your terminal. The relay is a **conduit, not a store**.



LIFECYCLE

The extension is a child of **gh copilot**. It opens the channel on launch, prints a pairing QR, and dies when you type **/exit**. The phone reports **"Session ended."**

TRUST BOUNDARY

Keys are generated at the endpoints and never transit the relay. Intercepted traffic is **ciphertext only**. No inbound ports, no tunnels, no browser auth on the phone.

— THE SURFACE

Reads like a real chat.

Rebuilt to mirror VS Code, GitHub Copilot and Claude-mobile: assistant on the left, you on the right, and tool calls folded inline — collapsed until you want them. Full phone width, vertical only, no horizontal scroll.

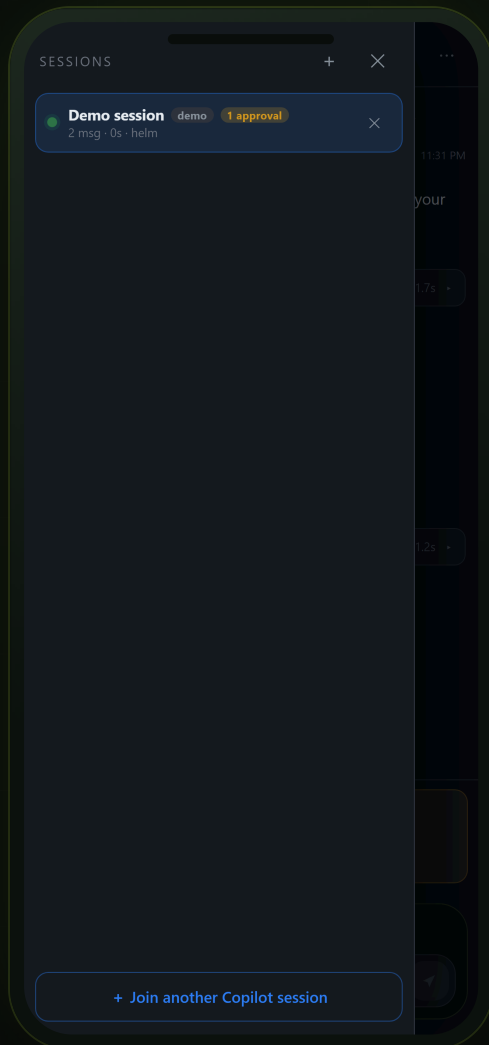


LIVE STREAM

TOOL, EXPANDED

APPROVE / DENY

- 1 Compact session header.** Title plus a live / quiet / ended dot — status at a glance, no wasted space.
- 2 Inline tool calls.** Folded into the flow, collapsed by default — the Anya / VS Code pattern.
- 3 Markdown, native.** Lists, bold, fenced code — rendered, not escaped.
- 4 Tap to expand.** Arguments and result (or error) in monospace, contained — never overflowing the page.
- 5 Space-smart composer.** One-row input that grows, with a mode pill and the working directory.
- 6 Hold-the-loop approvals.** The agent pauses on a file write; you Allow once, always, or Deny — from the couch.



— MANY TERMINALS, ONE APP

Every session, kept warm.

Join more than one **gh copilot** session and Helm keeps them all connected at once — each with its own live timeline held in memory. **Switch in a tap; scrollbar is preserved.**

Joined sessions persist to local storage and reconnect on launch. The drawer surfaces each one's status, message count, age and working directory — and flags any that are *waiting on you*.

— PER-SESSION STATE

**CONCURRENT**

Each session keeps its own **SecureChannel** and reducer — switching never drops a stream.

PERSISTENT

Stored under **helm.sessions.v1**, with migration from the legacy single-pairing key.

SELF-HEALING

A heartbeat watchdog eases **live** → **quiet** after silence, instead of falsely declaring the session dead.

— THE KIT

Signal-lime on terminal ink.

— PALETTE



— TYPE

FRAUNCES · DISPLAY

Helm keeps watch.

Editorial serif · opsz 9-144 · headlines & numerals

HANKEN GROTESK · BODY

Steer the agent remotely.

Humanist grotesk · 400 / 600 / 800 · running text

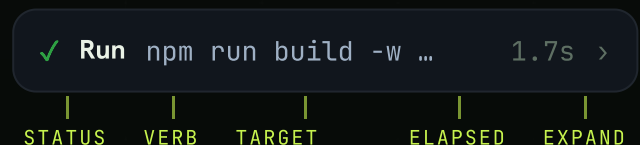
JETBRAINS MONO · SYSTEM

tool · 1.7s · ✓

Kickers · labels · code · elapsed times

HELM · PRODUCT DESIGN DOSSIER

— COMPONENT ANATOMY — TOOL CARD



Collapsed, a tool is a single quiet line:
outcome glyph, friendly verb, truncated
target, time. One tap reveals arguments and
result — detail on demand, never noise by
default.

— UNDER THE HOOD

Built to verify.

MOBILE	React 18 · Vite 6 · Capacitor 7	CRYPTO	ECDH · AES-256-GCM (WebCrypto)
RELAY	Supabase Realtime Broadcast	PAIRING	QR → on-device key derivation
HOST	@github/copilot-sdk extension	PERSISTENCE	localStorage · helm.sessions.v1
STATE	Vanilla store · useSyncExternalStore	DELIVERY	Netlify · usehelm.netlify.app

— ASSURANCE — PLAYWRIGHT, 412 PX

- ✓ Compact status bar shows a live session status
- ✓ Tool calls render inline & collapsed, then expand on tap
- ✓ Composer is space-smart (single row, not the legacy box)
- ✓ User prompts appear as a right-aligned bubble
- ✓ No horizontal scroll at phone width
- ✓ Session drawer opens and lists the joined session

6 passed · 15.9s — npm run test:e2e

— SECURITY MODEL

Air-gapped keys. Symmetric key born on the extension, scanned by QR.

Zero persistence. The relay stores nothing; the channel dies with the terminal.

No inbound surface. Outbound WebSocket only — no tunnels, ports, or phone-side auth.

NO DAEMONS

NO DB LOGS

CIPHERTEXT ONLY

 FIN

Ship from anywhere.

Helm — **Product Design Dossier**, Chat Surface v2. A secure mobile viewport for live **gh copilot** sessions: inline tool cards, right/left chat, multi-session, encrypted end to end.

Set in **Fraunces**, **Hanken Grotesk** & **JetBrains Mono**. Palette derived from the Helm signal-lime mark on terminal ink. Screens captured live at a 412 px device viewport.

SURFACE • [USEHELM.NETLIFY.APP](https://usehelm.netlify.app) SOURCE • [GITHUB.COM/AASIS21/HELM](https://github.com/aasis21/helm) © MMXXVI